AI-driven triage in emergency departments: A review of benefits, challenges and future directions

[DOI](#)

Emergency Departments face persistent overcrowding, limited resources and inconsistent patient prioritisation due to the subjective nature of traditional triage systems, especially under high-pressure conditions. The authors address this problem through a narrative review of peer-reviewed literature published between 2015 and 2024, sourced from databases including PubMed, Scopus, IEEE Xplore and Google Scholar. The review synthesises findings on AI-driven triage systems that use patient data such as vital signs and clinical information to support automated prioritisation. The reported outcome is that these AI systems generally improve triage accuracy, reduce waiting times, optimise resource allocation and enhance overall emergency department efficiency and patient care support. However, the authors explicitly note limitations, including data quality issues, algorithmic bias, lack of clinician trust and ethical concerns that currently hinder safe and equitable real-world deployment.

Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

[DOI](#)

Emergency departments struggle with triage efficiency, patient prioritization and resource allocation and while AI/ML holds promise in addressing these challenges, their actual impact on ED triage remains poorly characterized in the literature. The authors conducted a scoping review searching EMBASE, Ovid MEDLINE and Web of Science, screening 1,142 citations and selecting 29 peer-reviewed US primary research studies from 2013 - 2023 that used AI during triage and reported patient outcomes. Across those studies, ML models consistently outperformed conventional triage systems in predictive accuracy, disease identification, hospitalization prediction and resource allocation, including length-of-stay forecasting. The authors themselves acknowledge that title/abstract screening relied on subjective judgment, risking selection inconsistencies and that excluding high-bias studies may have omitted valuable insights, limiting the comprehensiveness and generalizability of their conclusions. Notably, the review raises red flags for inflated comparisons by pooling results across heterogeneous cohorts and metrics and never translates performance statistics into bedside consequences - a critical omission given the patient-safety stakes of ED triage.

Clinical Impact of Artificial Intelligence-Based Triage Systems in Emergency Departments: A Systematic Review

DOI

The clinical problem addressed was the growing pressure on emergency departments to improve triage accuracy and efficiency amid increasing patient volumes and limited resources. The method used was a systematic review conducted according to PRISMA 2020 guidelines, analyzing studies from major medical and engineering databases between 2020 and 2025 that evaluated AI-based emergency department triage systems, with risk of bias assessed using an adapted QUADAS-2 tool and findings synthesized narratively. The reported outcome was that AI systems showed promise in improving ED operations, with Voice-AI reducing documentation time by 19% and machine learning models lowering mis-triage rates by 0.3 - 8.9%, while also supporting faster triage and better clinical decision-making. The authors' stated limitations included undertriage risks, inconsistent accuracy across systems, reliance on predominantly single-center studies, workflow integration challenges, limited clinician acceptance data and a lack of evidence regarding patient-centered outcomes, equity and long-term impacts, highlighting the need for multi-center validation and standardized reporting.

Problem statement(Draft)

Emergency departments face overcrowding and inconsistent triage decisions driven by subjective scoring systems and variable clinician judgement. Recent AI triage models show improved predictive performance and efficiency. However, evidence is limited by poor generalisability, limited real-world integration and insufficient evaluation of patient-centred outcomes, safety (including under-triage risk) and bias. This 12-week pilot at Mercer General Hospital will assess an AI triage decision-support tool embedded in routine workflow, evaluating triage accuracy, time-to-decision and clinician acceptability.